

paper

PFR blueprint
Lemma 2.17

LaTeX statement

$$I[X; Y] \geq 0$$

qwen3-235b*

slogan

Mutual information $I[X; Y]$ is non-negative
for any random variables X and Y .

qwen3-8b*

$$\begin{bmatrix} -0.008 \\ 0.022 \\ 0.004 \\ \vdots \\ 0.011 \\ 0.019 \\ 0.022 \end{bmatrix}$$

embedding
4096-dim

Lean project

teorth/pfr
ProbabilityTheory

Lean source

```
theorem measureMutualInfo.nonneg :  
  0 ≤ measureMutualInfo μ
```

qwen3-235b*

slogan

For a probability measure μ ,
`measureMutualInfo` μ is non-negative.

qwen3-8b*

$$\begin{bmatrix} -0.014 \\ 0.019 \\ 0.007 \\ \vdots \\ 0.013 \\ 0.025 \\ 0.022 \end{bmatrix}$$

embedding
4096-dim

cosine sim
0.974
strong match
(sim ≥ 0.95)